

Tommaso Derossi

Data Scientist & Applied Researcher

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Summary

6+ years of experience in collaborating with the most diverse stakeholders, building data-driven systems (ETL → modeling → evaluation → serve/report) to translate messy data into rich information. Currently focused on research projects to advance manufacturing systems methodologies through classical and generative AI.

Domains: Research | Government | NGOs | Enterprises | Startups

Stack: Docker | AWS (S3/EC2/Lambda) | Spark | Python [Pytorch - Fastapi - Flask - MLflow - Metaflow - W&B - Airflow] | R | Java | Javascript

Experience

GEA Group - GEA Procomac S.P.A. November 2024 - Present

Industrial PhD
https://www.gea.com/en/

- Developing a data-driven recommendation system for industrial troubleshooting support.
- Vector databases, Bayesian statistics, clustering algorithms, agentic patterns, local LLMs.

Supervised Program for Alignment Research September 2025 - March 2026

Research Fellow Remote
https://sparai.org/

- AI Alignment research project on Chain of Thought (CoT) monitoring (supervised by Rauno Arike): development of CoT decomposition and attribution methods to study unsafe/strange LLM reasoning traces.

Freelance September 2024 - February 2026

AI Engineer

- Developed a recommendation/ranking system for job-candidate matching.
- LLM evaluations, CI/CD practices

Tomorrow S.R.L. February 2023 - July 2024

Data Scientist
https://www.tomorrow srl.com/

- Conceptualized and developed data transformation pipelines integrated with machine learning models (Bayesian hierarchical models, Random forest, Gradient boosting) for the analysis and presentation of IOT data (Predictive maintenance, KPIs optimization, Data-driven actionable insights, Time series forecasting)
- Designed and implemented customized AI conversational agent for purchasing assistance based on LLMs (LLMs prompting and finetuning, RAG, LLamaIndex, Langhchain)
- Prototyped a recommendation system and visualization dashboard for data-driven soccer scouting decision-making support, based on wide high frequency match-events data and machine learning models (Pytorch, Sklearn)
- Created marketing automation pipelines from target audience selection and content production to campaign creation and management with human-in-the-loop for different advertisement platforms (Linkedin, Meta, X)

WATERONMARS S.R.L. March 2022 - February 2023

Data Scientist
https://www.wateronmars.it/

- Built ETL pipelines (Visokio Omniscope) for the download (Web-scraping, API) and analysis of web content, in particular, social media data (textual data, images, videos, geospatial and temporal data)
- Implemented and integrated predictive models for data enrichment in the domains of computer vision (OpenCV, SimpleCV) and natural language processing (Spacy, Transformers, NLTK, Gensim, TextBlob, Prodigy)
- Created highly customized data visualization dashboards for various clients and types of analysis (Geospatial analytics - GIS, Election campaign analysis, Brand monitoring, Buyer persona, USP analysis, E-commerce analytics)

OMIGRADE S.R.L. March 2019 - October 2020

Machine Learning engineer
https://www.omigrate.it/

- Developed an LSTM model for an interactive classification web application; engineered distributed retraining/monitoring pipeline on Hadoop with Spark RDDs and PySpark-style processing.
- MLOps: standardized data prep, periodic retraining, monitoring, performance analysis.

Education

Università degli studi di Parma November 2024 - Present

Artificial Intelligence PhD Student

- Research focus on AI applications in industrial troubleshooting systems, time series analysis, Large Language Models safety and interpretability.

Università degli studi di Torino September 2019 - April 2023

Stochastics and Data Science Master Degree
https://www.unito.it/

- Degree course in Mathematics fully taught in English

Thesis: "Modelling the discourse around nature in Twitter in presence of highly imbalanced data."

Thesis topics: Natural Language Processing, Transformers models, Network Science, Filtering Techniques, Sentiment Classification, Semantic Clustering, Topic Modeling

KU Leuven September 2020 - July 2021

Statistics 1 year exchange programme
https://www.kuleuven.be/

- Second year of master’s degree spent in Belgium at KU Leuven. Some of the main courses attended (10 courses in total): Optimization and Numerical Methods (17/20), Concepts of Bayesian Data Analysis (16/20), Data Mining and Neural Networks (15/20), and Modern Data Analytics (15/20)
- Participated in Data4Good Challenge (Emergent Leuven)

Università degli studi di Parma September 2016 - September 2019

Physics Bachelor Degree
https://www.unipr.it/

- Awarded for being the top-ranked student in credits earned relative to grade point average during the second academic year
- Thesis: "Methodological aspects of neural networks integrated within an application for the classification of relational database columns."
- Thesis topics: Artificial Neural Networks, Deep Recurrent Neural Networks (LSTM), Distributed Model Training

Liceo Scientifico "Giacomo Ulivi" 2011 - 2016

Scientific Oriented High School

South African College High School 2012

English School Attendance, 3 months

Skills			
Python	R	MATLAB	SQL, NoSQL
MS Office (Excel), LaTeX	ML/DL (Pytorch, Tensorflow)	Spark, PySpark	Visokio Omniscope
Docker, Git	AWS (S3, EC2, Lambda)	LLM Evaluation	ML Flow, Weights&Biases, Metaflow, Airflow

Languages		
Italian	English	Spanish
Mother tongue	Fluent	Intermediate